

MECH 467 – Computer Control of Mechatronics Systems
Prof. Y. Altintas
The University of British Columbia, Department of Mechanical Engineering

Project # III: Simulation of Contouring Performance in Coordinated Two Axis Motion
Due date: December 2, 2013

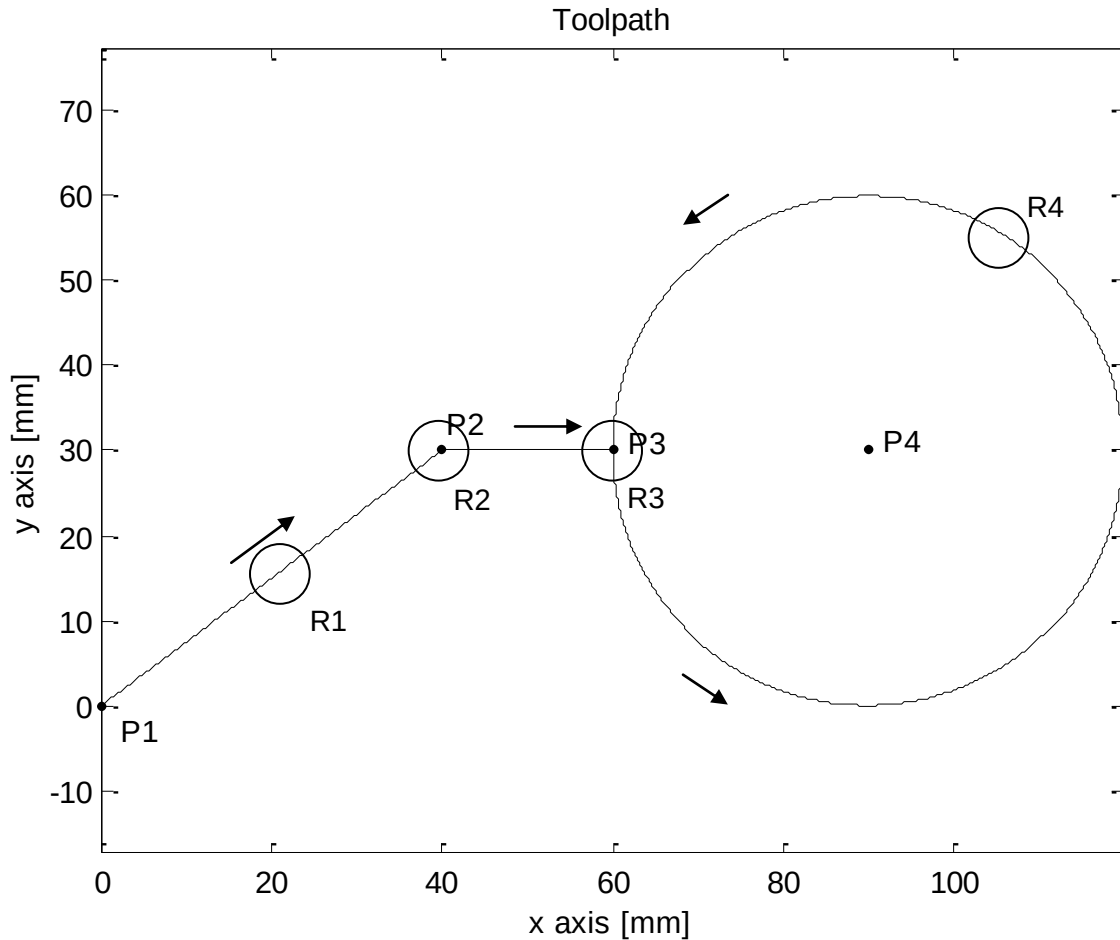


Figure 1: Reference Toolpath to be used in Contouring Performance Evaluation.

A sample toolpath is shown in Figure 1, comprising of two linear segments and a full circle. The coordinates of the segment end points are $P1(0,0)$ [mm], $P2(40,30)$ [mm], and $P3(60,30)$ [mm]. The center of the circle is $P4(90,30)$ [mm].

Part A – Trajectory Generation

A.1. Using trapezoidal feedrate profiling, generate the axis commands as functions in discrete-time ($x = x(k \cdot T_s)$, $y = y(k \cdot T_s)$, where $k = 0, 1, \dots$). Use a sampling period of $T_s = 1$ [msec]. Assume that the desired feedrate value is $F = 200$ [mm/sec], and the tangential acceleration and deceleration values are $A = -D = 1000$ [mm/sec²]. The feed profile starts from rest at $P1$, and also comes to full stops at the corners $P2$ and $P3$. The final feedrate, after performing the counter-clockwise circular interpolation, is also zero. Plot the following:

- i. The generated toolpath: x axis position commands versus y axis position commands (x_r vs. y_r),

- ii. The displacement (s), feedrate ($\dot{s}$), and tangential acceleration ($\ddot{s}$) profiles as functions of time ($t = k \cdot T_s$),
- iii. The axis position (x_r, y_r), velocity ($\dot{x}_r, \dot{y}_r$), and acceleration ($\ddot{x}_r, \ddot{y}_r$) commands as functions of time ($t = k \cdot T_s$).
- iv. Please make sure that related plots are kept together in an organized engineering report.

[Note: Graduate students must use Jerk limited trajectory generation profile with $J=50,000$ [mm/sec³]

Part B – Two Axis Controller Design

In this project, a linear motor driven XY table will be modeled, and its response will simulated. The experimentally identified drive dynamics of the XY table are given as:

Axis Parameters	X Axis	Y Axis
Amplifier Gain (K_a [A/V])	0.28568	0.2802
Motor Force Constant (K_t [N/A])	99.5447	135.1173
Encoder Gain (K_e [mm/m])	1000	1000
Mass (m [kg])	5.447	31.482
Friction (b [N/m/sec])	18.0928	124.9376

Table 1: Axis Parameters

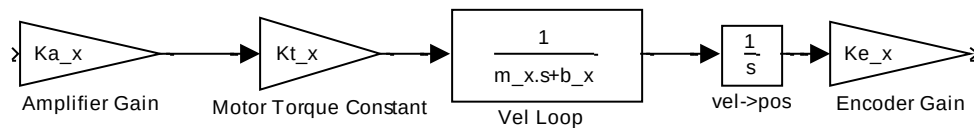


Figure 2: Linear Motor Driven Axis Model(X Axis)

B.1. The design methodology for Integrator added Lead-Lag controller is introduced in the previous laboratory project. Using those design steps, design 3 different Integral-Lead-Lag position controllers for both X and the Y axes. The design requirements for the position controllers are given in Table 2. (Use $K_i = \omega_c/10$)

	Cross Over Frequency(ω_c [hz]) and Phase margin (ϕ [deg])
Case 1(Low Bandwidth)	X Axis $\omega_c=20$ [hz], $\phi = 60$ [deg] Y Axis $\omega_c=20$ [hz], $\phi = 60$ [deg]
Case 2(High Bandwidth)	X Axis $\omega_c=40$ [hz], $\phi = 60$ [deg] Y Axis $\omega_c=40$ [hz], $\phi = 60$ [deg]
Case 3(Mismatched Dynamics)	X Axis $\omega_c=40$ [hz], $\phi = 60$ [deg] Y Axis $\omega_c=20$ [hz], $\phi = 60$ [deg]

Table 2: Closed loop pole locations

- i. Determine poles and zeros, and obtain the bandwidth of the closed loop system for different cross-over frequencies of each axis.
- ii. Compare the bandwidth as well as the transient closed loop responses of each axis for each case (the overshoot and rise times in time domain). Tabulate your results!
- iii. Transform the controller into discrete-time using the Euler's approximation $s \approx (z - 1)/(T_s z)$ and derive the z-domain transfer function.

(Note: The students, who wish to obtain extra 5% mark on the final grade, can design a pole placement controller for this table; test it on Virtual CNC and also test it in the laboratory. It will be a great experience and you will learn the full engineering steps in designing a multi-axis controller for machines. If you can't test it on the XY Table in the lab, testing in Virtual CNC is okay with 1 point less extra mark. It is compulsory for graduate students.)

Part C – Contouring Performance Simulation

C.1. Implement the discrete position controller with the continuous velocity dynamics using a zero-order hold block in Matlab/Simulink environment. Apply the generated reference trajectory in A.1 to the two-axis position control system, constructed as shown in Figure 3.

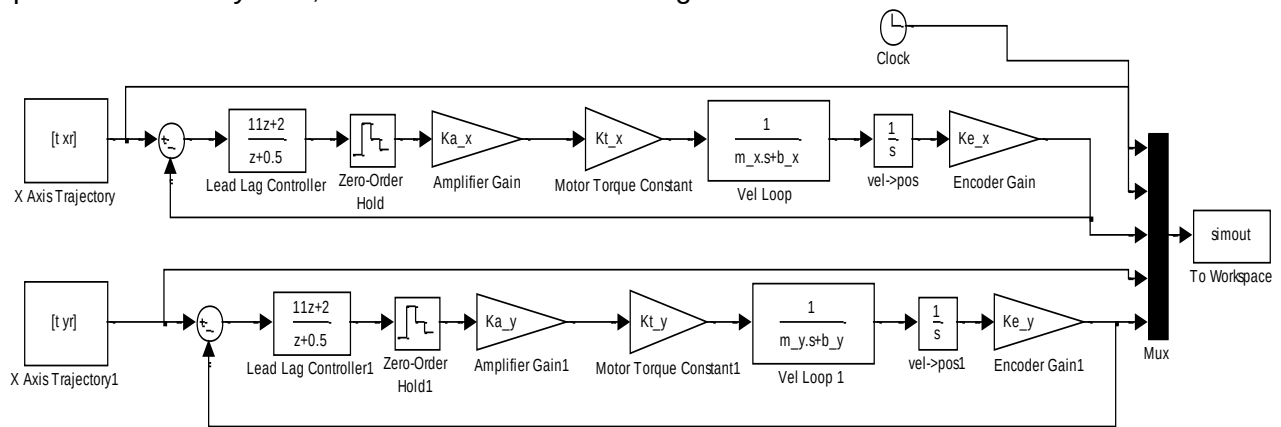


Figure 3: Simulink Model for Trajectory Simulation

(Hint: Please use **From Workspace** block from **Sources** menu in the Simulink Library to read your trajectory data (t [sec], x_r [mm] and y_r [mm]) existing on the Matlab's workspace. Furthermore, use the **To Workspace** block from the **Sinks** menu to import back your simulation results into the **Matlab Workspace** for evaluation. A **mux** block will also help to stack simulation arrays together)

Simulate the predicted tool motion, and overlay the results on top of the reference toolpath (x_r vs. y_r) and simulated axis trajectory (x vs. t and y vs. t) graphs. Also compute and plot the tracking error profiles in the x and y axes, ($e_x = x_r - x$, $e_y = y_r - y$) as functions of time (e_x vs. t and e_y vs. t).

Perform the following simulations for the 3 different cases given in Part B.

C.2. Comment on the tracking performance throughout the motion, and also on the (geometric) contouring accuracy at the critical regions R1, R2, R3, and R4 shown with circles in Figure 1. Zoom into these regions and approximately measure the largest value of the predicted contour error. What is the contour error value in region R1 during constant feedrate motion? What are the corresponding axis tracking errors in this region? Elaborate on your observations.

C.3. Re-generate the reference trajectory, using a desired feedrate of $F = 250$ [mm/sec]. Plot the new displacement (s), feedrate ($\dot{s}$), tangential acceleration ($\ddot{s}$) profiles, as well as axis position (x_r, y_r), velocity ($\dot{x}_r, \dot{y}_r$), and acceleration ($\ddot{x}_r, \ddot{y}_r$) commands, all as functions of time (t). Re-simulate the two-axis response to the new trajectory. Similar to the steps carried out in C1 and C2, overlay the simulation results on top of the toolpath and axis trajectory commands. Plot the tracking error history and zoom into the critical regions R1, R2, R3, and R4 in the predicted toolpath. How do the results compare with those obtained in C2?

C.4. In Case 3, the cross-over frequencies of the X and Y axes are assigned differently. Hence, the two axes will no longer have the identical position loop dynamics. Their bandwidths are different. Simulate the contouring performance of the *mismatched* axis dynamics for the high speed reference trajectory. What can you say about the tracking and contouring performance in this case? Elaborate on your findings in relation to achieving high contouring accuracy in multi-axis positioning systems.

C.5. This section is for the experimental implementation of **your own trajectory on the real XY table**. A picture of the actual Linear Motor driven XY table is given in Figure 4.



Figure 4: Linear Motor Driven XY Table

The maximum workspace for the XY table is $18[mm] \times 18[mm]$. Define a geometry you prefer using **only** linear and circular interpolation techniques. The geometry can be any shape as long as it contains lines and circles, but does not exceed the boundaries of the workspace. Interpolate your geometry using $F = 100[mm/sec]$ and $A=D = 250 [mm/sec^2]$ at $T_s=1[msec]$ sampling time. Prepare array of position commands for the X and Y axes where $x_{r_{1 \times m}}$ is the column reference position command array for the X axis, and $y_{r_{1 \times m}}$ is for the Y Axis. For the position controller, you will be using the controller that you designed for the Case 1. Save the z-domain transfer function as well as the X,Y axis position command arrays in a **.mat** file in MATLAB, and bring them in a memory stick. You will also implement your own controller into the experimental system. So, test and make it sure that your entire trajectory works in closed loop!!!(use your simulation scheme in part C to test it)

Notes: Once you set up the system in Simulink, all answers can be automatically plotted in MATLAB. The questions have lengthy explanations, but the work is not extensive.

Report Requirements:

The report must be typed and orderly. It accounts for 70% of your lab grade. Poorly presented reports will not be accepted. Equations may be handwritten into the report, but they must be written very neatly and numbered. Please be as concise as possible. Do not provide unnecessary information. Credit will however be given for concise comments which demonstrate a good understanding of the material covered in the lab.

The format of the report is.

- 1. Title Page:** Your name, student ID, email, etc
- 2. Abstract:** 3-4 lines describing the objective of the laboratory as you understand it.
- 3. Introduction:** Describe the role of multi-axis trajectory design in industry and the layout of your report (maximum 5 to 6 lines)

4. **Body:** In this Part you need to present the results in all questions clearly but concisely with short discussions. Laboratory result must also be presented.
5. **Conclusion:** Explain what you have understood from the lab (5-8 lines)